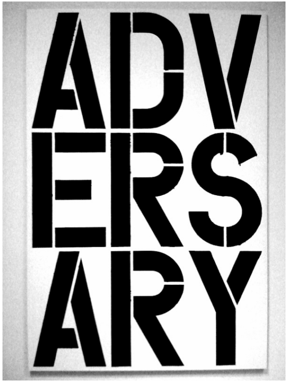
Adversarial Bandits — EWRL 2026



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- The train-test model of statistical learning is not well suited for learning when new data is being generated at a fast pace
- Online learning algorithms incrementally adjust their models as new data arrive

Some history



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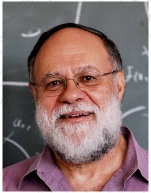
- Volodya Vovk independently develops a related framework (Aggregating strategies, 1990)
- Similar ideas also independently emerged in game theory and information theory:
  - Tom Cover
  - Rakesh Vohra and Dean Foster
  - Sergiu Hart and Andreu Mas-Colell





Prediction with expert advice

A sequential decision-making problem

- d actions
- Unknown deterministic assignment of losses to actions $\boldsymbol{\ell}_t = (\ell_t(1), \dots, \ell_t(d)) \in [0,1]^d$ for each time step $t = 1, 2, \dots$

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.7 .3 .2 .4 .1 .6 .7 .4 .9 .2

Regret

$$R_T = \underbrace{\mathbb{E}\left[\sum_{t=1}^{T} \ell_t(I_t)\right]}_{\text{total loss of player}} - \min_{i=1,\dots,d} \underbrace{\sum_{t=1}^{T} \ell_t(i)}_{\text{total loss of best action}}$$

- Expectation is over player's randomization

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- Expectation is over player's randomization
- Online linear optimization in the simplex Δ_d

Lower bound for the simplex

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&= \mathbb{E}\left[\max_{i=1,\ldots,d} \sum_{t=1}^{T} \left(\frac{1}{2} - \ell_t(i)\right)\right] \\
&= \frac{1}{2}\,\mathbb{E}\left[\max_{i=1,\ldots,d} \sum_{t=1}^{T} \varepsilon_t(i)\right]
\end{aligned}
$$

where $\varepsilon_t(i) \in \{-1, 1\}$ are uniform i.i.d. Rademacher random variables

Lower bound for the simplex: finishing up

- A standard result on Rademacher sums:

$$\mathbb{E}\left[\max_{i=1,\dots,d}\sum_{t=1}^{T}\varepsilon_t(i)\right] = \big(1-o(1)\big)\sqrt{2T\ln d}$$

Lower bound for the simplex: finishing up

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- Therefore, $R_T = \frac{1}{2}\,\mathbb{E}\left[\max_{i=1,\ldots,d} \sum_{t=1}^{T} \varepsilon_t(i)\right] = \boxed{\big(1 - o(1)\big)\sqrt{\frac{T}{2}\ln d}}$

Exponentially weighted forecaster (Hedge)

- At time $t+1$ the Hedge algorithm draws action I_{t+1} from $\boldsymbol{p}_{t+1}$, where
$$p_{t+1}(i) = \frac{\exp\left(-\eta\sum_{s=1}^{t}\ell_s(i)\right)}{\sum_{j=1}^{d}\exp\left(-\eta\sum_{s=1}^{t}\ell_s(j)\right)}$$

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- This is an instance of Follow-The-Regularized-Leader (FTRL) algorithm

$$\boldsymbol{p}_{t+1} = \operatorname*{argmin}_{\boldsymbol{p}\in\Delta_d} \psi(\boldsymbol{p}) + \sum_{s=1}^{t} \boldsymbol{p}^\top \boldsymbol{\ell}_s$$

with entropic regularizer $\psi(\boldsymbol{p}) = \dfrac{1}{\eta}\sum_i p(i)\ln p(i)$ for $\boldsymbol{p} \in \Delta_d$

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- This matches the asymptotic lower bound, including constants
- We prove this later in a more general setting

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2. Player gets feedback information: Only $\ell_t(I_t)$ is revealed

? .3 ? ? ? ? ? ? ? ?

Hedge revisited

Player's strategy must use loss estimates $\ell_s(i)$ instead of true losses $\ell_s(i)$

- $p_t(i) \quad \propto \quad \exp\left(-\eta \sum_{s=1}^{t-1} \widehat{\ell}_s(i)\right) \qquad i = 1, \ldots, d$

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- The learning rare η balances these two components

Key properties of the estimator

- Unbiasedness

$$\mathbb{E}_t\left[\widehat{\ell}_t(i)\right] = \frac{\ell_t(i)}{\mathbb{P}_t\big(\ell_t(i) \text{ observed}\big)} \times \mathbb{P}_t\big(\ell_t(i) \text{ observed}\big) + 0 = \ell_t(i)$$

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- Second-moment bound

$$\mathbb{E}_t\left[\widehat{\ell}_t(i)^2\right] = \frac{\ell_t(i)^2}{\mathbb{P}_t(\ell_t(i) \text{ observed})^2} \times \mathbb{P}_t(\ell_t(i) \text{ observed}) + 0$$

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&\le \frac{1}{\mathbb{P}_t\big(\ell_t(i) \text{ observed}\big)}
\end{aligned}$$

Regret analysis

$$\frac{W_{t+1}}{W_t} = \sum_{i=1}^{d} \frac{w_{t+1}(i)}{W_t} \qquad p_t(i) = \frac{1}{W_t} \exp\left(-\eta \sum_{s=1}^{t-1} \widehat{\ell}_s(i)\right) = \frac{w_t(i)}{W_t}$$

is a r.v.!

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$$\leq \sum_{i=1}^{d} p_t(i) \left(1 - \eta\, \widehat{\ell}_t(i) + \frac{\big(\eta\, \widehat{\ell}_t(i)\big)^2}{2}\right)$$

(using $e^{-x} \leq 1 - x + x^2/2$ for all $x \geq 0$)

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Regret analysis (cont.)

Taking logs, using $\displaystyle\sum_{t=1}^{T} \ln \frac{W_{t+1}}{W_t} = \ln \frac{W_{T+1}}{W_1}$ and $\ln(1+x) \le x$ yields

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For the best action $k = \operatorname*{argmin}_{i=1,\dots,d} \sum_{t=1}^{T} \ell_t(i)$, using $W_1 = d$ and $w_{T+1}(k) = e^{-\eta \sum_t \widehat{\ell}_t(k)}$

$$\ln \frac{W_{T+1}}{W_1} \ge \ln \frac{w_{T+1}(k)}{W_1} = -\eta \sum_{t=1}^{T} \widehat{\ell}_t(k) - \ln d$$

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Putting together and dividing both sides by $\eta > 0$ gives

$$\sum_{t=1}^{T} \sum_{i=1}^{d} p_t(i)\widehat{\ell}_t(i) - \sum_{t=1}^{T} \widehat{\ell}_t(k) \le \frac{\ln d}{\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \sum_{i=1}^{d} p_t(i)\widehat{\ell}_t(i)^2$$

Regret analysis (cont.)

Recall where we were:

$$\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\widehat{\ell}_t(i) - \sum_{t=1}^{T}\widehat{\ell}_t(k) \le \frac{\ln d}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\widehat{\ell}_t(i)^2$$

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Take expectation w.r.t. $I_1, \dots, I_T$

$$\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\mathbb{E}_t\big[\widehat{\ell}_t(i)\big] - \sum_{t=1}^{T}\mathbb{E}_t\big[\widehat{\ell}_t(k)\big]\right] \le \frac{\ln d}{\eta} + \frac{\eta}{2}\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\mathbb{E}_t\big[\widehat{\ell}_t(i)^2\big]\right]$$

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Take expectation w.r.t. $I_1,\ldots,I_T$

$$\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\mathbb{E}_t\left[\widehat{\ell}_t(i)\right] - \sum_{t=1}^{T}\mathbb{E}_t\left[\widehat{\ell}_t(k)\right]\right] \le \frac{\ln d}{\eta} + \frac{\eta}{2}\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\mathbb{E}_t\left[\widehat{\ell}_t(i)^2\right]\right]$$

Loss estimates are unbiased:

$$\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\ell_t(i)\right] - \sum_{t=1}^{T}\ell_t(k) \le \frac{\ln d}{\eta} + \frac{\eta}{2}\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\mathbb{E}_t\left[\widehat{\ell}_t(i)^2\right]\right]$$

Regret analysis (cont.)

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The left-hand side is just the regret R_T

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Regret analysis (cont.)

$$R_T \le \frac{\ln d}{\eta} + \frac{\eta}{2}\,\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d} p_t(i)\mathbb{E}_t\left[\widehat{\ell}_t(i)^2\right]\right]$$

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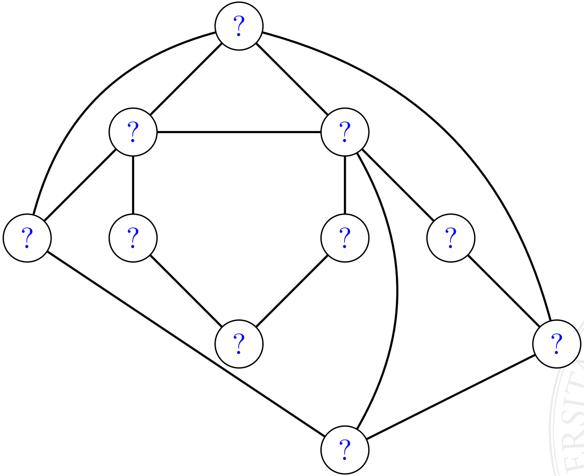
\end{aligned}
$$

We now analyze a setting in which the the probability of observing the loss interpolates between experts and bandits

A feedback graph over actions



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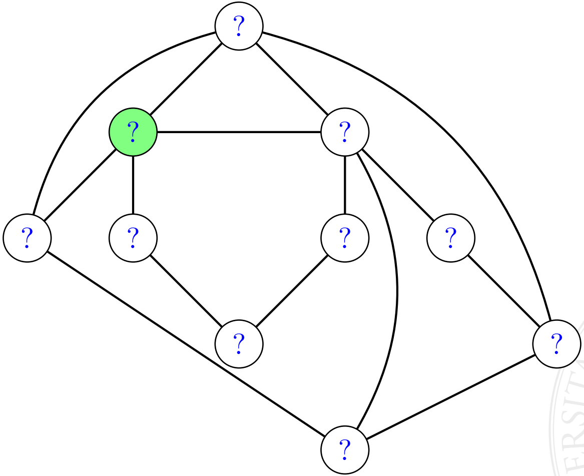

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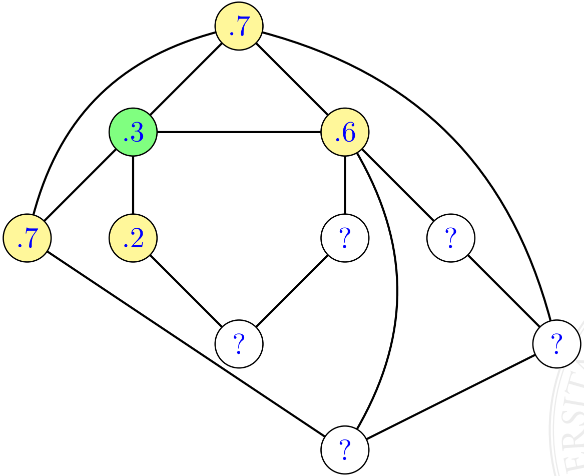
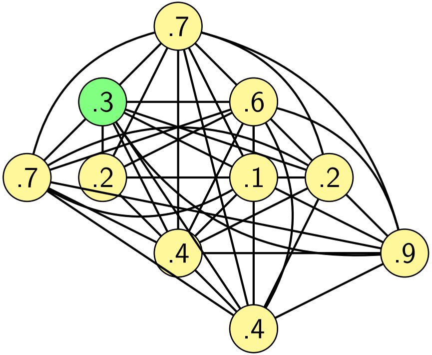


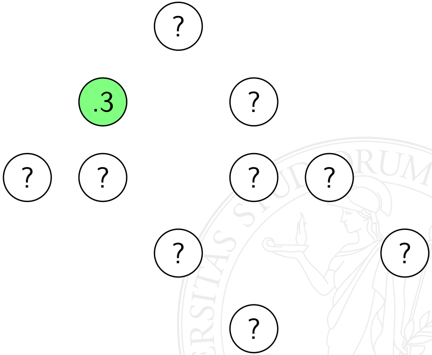
$\ell_t(i)$ is observed iff $I_t \in \{i\} \cup \mathcal{N}_G(i)$

Recovering expert and bandit settings

Experts: clique



Bandits: edgeless graph



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$$R_T \leq \frac{\ln d}{\eta} + \frac{\eta}{2}\,\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{d}\frac{\mathbb{P}_t(i \text{ is played})}{\mathbb{P}_t\big(\ell_t(i) \text{ is observed}\big)}\right] \qquad \text{(variance bound)}$$

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$$\le \frac{\ln d}{\eta} + \frac{\eta}{2} T\,\alpha(G) \qquad \text{(Caro-Wei Lemma)}$$

$\alpha(G)$ is the independence number of G

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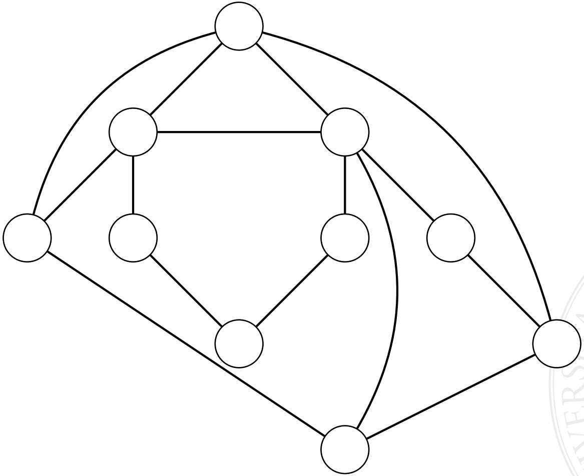
The size of the largest independent set in G



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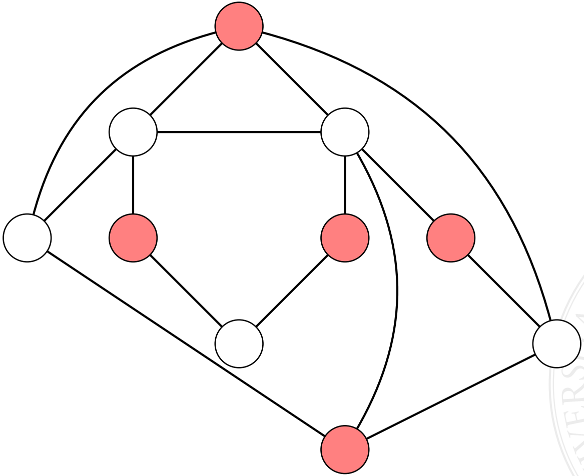

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Special cases

Experts (clique): $\alpha(G) = 1$ $\quad R_T \le \sqrt{T \ln d}$ Hedge algorithm

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Better bound for bandits $\sqrt{dT}$ achieved by FTRL with Tsallis $\frac{1}{2}$-entropy regularizer

$$\psi(\boldsymbol{p}) = 2\left(1 - \sum_{i=1}^{d} \sqrt{p(i)}\right)$$

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- The resulting regret is at least $\varepsilon T = \sqrt{dT}$

Bandit lower bound (cont.)

$N_T(i)$ is number of times action i is chosen by a deterministic player

$$\mathbb{E}[R_T] = \underbrace{\frac{1}{d}\sum_{i=1}^{d}}_{\substack{\text{random draw} \\ \text{of good action}}} \underbrace{\sum_{j \neq i} \varepsilon\, \mathbb{E}_i\big[N_T(j)\big]}_{\substack{\text{regret for not playing} \\ \text{the good action}}} = \varepsilon \left(T - \frac{1}{d}\sum_{i=1}^{d} \mathbb{E}_i\big[N_T(i)\big] \right)$$

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- This key inequality establishes that the good arm does not get played significantly more often than any random arm

Bandit lower bound (cont.)

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- Choosing ε of order $\sqrt{d/T}$ gives $\mathbb{E}[R_T] = \Omega(\sqrt{dT})$

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- Vanilla FTLR has linear regret with switching cost

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- Choosing $B = (T/d)^{1/3}$ gives $R_T = \mathcal{O}\left(d^{1/3}T^{2/3}\right)$

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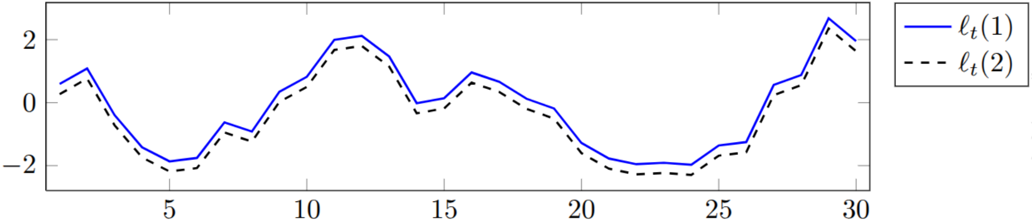

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